

HOMWORK 2 SOLUTIONS

Problem 1

Use the provided dataset "HW2_data.csv" to build a multiple linear regression model

$$y = X\beta + \sigma\epsilon,$$

Where X is an $n \times p$ matrix. The size of the sample is $n = 50$ and the number of predictor variables is $p = 3$. Note that the column of 1's for the intercept term is already included in the dataset.

You can use programming software (R, Python, etc.) to do the computations. You do not need to submit the code but please show your work, including equations, test statistics and conclusions.

- (1) (10 pts) Find the estimator $\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3)$ and the distribution of $\hat{\beta}_3$.

Solution:

$$\hat{\beta} = (X'X)^{-1}X'y = (-0.510, -0.534, 0.087)$$

$$E(\hat{\beta}) = \beta \quad Cov(\hat{\beta}) = \sigma^2(X'X)^{-1}$$

$\hat{\beta}_3$ is an unbiased estimator of the parameter β_3 . Its mean is the true value of β_3 , and its variance is the third diagonal element of the covariance matrix $\sigma^2(X'X)^{-1}$. Note that we don't replace β_3 and σ^2 with the values of the estimates $\hat{\beta}_3$ and $\hat{\sigma}^2$. Instead, we keep them as the unknown parameters when we are talking about the distribution of $\hat{\beta}_3$, so

$$\hat{\beta}_3 \sim N(\beta_3, 0.022\sigma^2)$$

- (2) (10 pts) Perform an F -test for the hypothesis $H_0 : \beta_3 = 0$ with the $\alpha = 0.01$ level of significance. Provide the value of the F test statistic and the corresponding critical value.

Solution:

Fit the reduced linear model only use β_1 and β_2 , and compute the reduced residual sum of squares.

$$H_{red} = X_{red}(X'_{red}X_{red})^{-1}X'_{red} \quad \hat{e}'_{red}\hat{e}_{red} = y'(I - H_{red})y$$

Compute the residual sum of squares with the full model.

$$H = X(X'X)^{-1}X' \quad \hat{e}'\hat{e} = y'(I - H)y$$

$$F_{p-q, n-p} = \frac{(\hat{e}'_{red}\hat{e}_{red} - \hat{e}'\hat{e})/(p-q)}{\hat{e}'\hat{e}/(n-p)} = 6.040 < F_{0.01, 1, 47} = 7.207$$

So, we fail to reject the H_0

- (3) (10 pts) Use the marginal distribution of $\hat{\beta}_3$ to conduct a t -test for the hypothesis $H_0 : \beta_3 = 0$ with the $\alpha = 0.01$ level of significance. Provide the value of the test statistic and the p value.

Solution:

$$\hat{\beta}_3 \sim N(\beta_3, 0.022\sigma^2) \quad \hat{\sigma}^2 = \frac{1}{n-p} \hat{e}'\hat{e} = 0.058$$

$$t_{n-p} = \frac{\hat{\beta}_3}{\sqrt{0.022\hat{\sigma}^2}} = 2.458 \quad p = 0.018 > 0.01$$

So, we fail to reject the H_0

- (4) (10 pts) Compute the leverage score for the first and second observations. Which data point is more influential? What is the issue arise with a very high leverage value.

Solution:

$$h_{11} = 0.037 \quad h_{22} = 0.032$$

A higher leverage score suggests the point is influential in that the variance of the residual is small and hence the fitted curve will pass through close to the point regardless of other observations.

- (5) (10 pts) Given a new data point $x = (1, 0.12, 0.56)$, find the predicted y value and its 95% confidence interval.

Solution:

$$y_{pred} = x'\hat{\beta} = -0.525, \quad t_{\alpha/2, n-p} = 2.012$$

The confidence interval is

$$\left(y_{pred} - t_{.025, 47} \sqrt{\hat{\sigma}^2 x'(X'X)^{-1}x}, \quad y_{pred} + t_{.025, 47} \sqrt{\hat{\sigma}^2 x'(X'X)^{-1}x} \right)$$

which is evaluated as $(-0.600, -0.450)$.

Problem 2

Consider the p dimensional multiple linear model

$$y = X\beta + \sigma\varepsilon$$

where $X'X = D$ is a diagonal matrix, $D = \text{diag}(d_j)$, and one of these elements, say $d_1 = \eta$ is very small and $1/\eta$ is very large.

- (1) (10 pts) What is the problem with using the ordinary least squares estimate; i.e.

$$\hat{\beta} = (X'X)^{-1}X'y?$$

Solution: Recall that $\hat{\beta}_j \sim N(\beta_j, \sigma^2 C_j)$, where C_j is the j th diagonal element of $(X'X)^{-1}$, so if C_j is big, there is a large variance for β_j and the estimator could be unreliable. In this case, $C_1 = 1/\eta$ is very large, so the OLS estimator for β_1 is unreliable.

- (2) (10 pts) A ridge estimator is proposed; i.e.

$$\hat{\beta}_R = (X'X + \lambda I)^{-1} X'y.$$

How does this solve the problem alluded to in part (1). In particular, find the variance of the ridge estimator for β_1 .

Solution:

$$\text{Var}(\hat{\beta}_{R1}) = \sigma^2 \frac{\eta}{(\eta + \lambda)^2}$$

For a positive λ value, $\eta/(\eta + \lambda)^2 < 1/\eta$, so the variance of the ridge estimator will be much smaller than the OLS estimator, which makes the ridge estimator more reliable.

- (3) (10 pts) What is the bias of the ridge estimator; i.e.

$$E(\hat{\beta}_R) - \beta.$$

Solution:

$$E(\hat{\beta}_R) - \beta = D(D + \lambda I)^{-1}\beta - \beta = -\lambda(D + \lambda I)^{-1}\beta$$

Or

$$E(\hat{\beta}_{Rj}) - \beta_j = \beta_j \frac{d_j}{d_j + \lambda} - \beta_j = -\beta_j \frac{\lambda}{d_j + \lambda}$$

- (4) (10 pts) When choosing a value for λ , why is it important not to set the value to be either too large or too small.

Solution: This is about a bias-variance trade-off. If λ is too large, then it will shrink the ridge estimator to 0 and the bias will be too large. If λ is too small, then it cannot effectively reduce the variance of $\hat{\beta}$.

- (5) (10 pts) For what values of λ , in terms of η , σ and the true value of β_1 , is it that the MSE of $\hat{\beta}_{R1}$ is smaller than that of the corresponding value of $\hat{\beta}_1$.

Solution:

$$\begin{aligned} \text{MSE}(\hat{\beta}_{R1}) &= E[(\hat{\beta}_{R1} - \beta_1)^2] = E[(\hat{\beta}_{R1} - E[\hat{\beta}_{R1}])^2] + (E[\hat{\beta}_{R1}] - \beta_1)^2 \\ &= \text{Var}(\hat{\beta}_{R1}) + \text{Bias}(\hat{\beta}_{R1})^2 = \sigma^2 \frac{\eta}{(\eta + \lambda)^2} + \beta_1^2 \frac{\lambda^2}{(\eta + \lambda)^2} \end{aligned}$$

$$\text{MSE}(\hat{\beta}_1) = \text{Var}(\hat{\beta}_1) = \sigma^2/\eta$$

Solve the inequality,

$$\begin{aligned} \sigma^2 \frac{\eta}{(\eta + \lambda)^2} + \beta_1^2 \frac{\lambda^2}{(\eta + \lambda)^2} &< \frac{\sigma^2}{\eta} \quad \Leftrightarrow \quad \sigma^2 \eta + \beta_1^2 \lambda^2 < \frac{\sigma^2}{\eta} (\eta + \lambda)^2 \\ \Leftrightarrow \quad \beta_1^2 \lambda &< \frac{\sigma^2}{\eta} \lambda + 2\sigma^2 \quad \Leftrightarrow \quad (\beta_1^2 - \frac{\sigma^2}{\eta}) \lambda < 2\sigma^2 \end{aligned}$$

Therefore, we need to have $\lambda < 2\sigma^2(\beta_1^2 - \sigma^2/\eta)^{-1}$ when $(\beta_1^2 - \sigma^2/\eta)$ is positive.

If $(\beta_1^2 - \sigma^2/\eta)$ is negative, we only need to have $\lambda > 0$.